Additional file 3. Results of postoperative ROM.

| Outcome | Study ID | Findings | | P value |
|---|---|---|---|---|
| | | FSA+LCP | LCP | |
| Forward elevation (degree) | Cui 2019 | 144.04±21.37 | 128.49 ± 22.81 | 0.010 |
| | Davids 2020 | 142.3（110-160） | 140.3（70-180） | 1.000 |
| | Kim 2020 | 127.50 ± 18.74 | 109.00 ± 14.10 | 0.001 |
| | Lee 2019 | 148.9 ± 19.8 | 125.3 ± 21.4 | 0.042 |
| Abudtion (degree) | Cui 2019 | 140.64 ± 20.34 | 122.37 ± 22.31 | 0.002 |
| | Davids 2020 | 125(90-160) | 120.6(70-170） | 0.800 |
| | Kim 2020 | 118.25 ± 17.49 | 48.25 ± 17.71 | 0.000 |
| External rotation (degree) | Cui 2019 | 58.96 ± 8.49° | 55.09 ± 8.63° | 0.090 |
| | Davids 2020 | 54.2（20-80） | 51.5（5-80） | 0.800 |
| | Kim 2020 | 38.75 ± 7.58 | 34.00 ± 14.74 | 0.341 |
| | Lee 2019 | 58.7 ± 18.3 | 53.2 ± 16.7 | 0.175 |
| Internal rotation (degree) | Cui 2019 | T12 level (L5-T5) | L1(buttock-T5) | 0.438 |
| | Kim 2020 | L5 | L1 | 0.000 |